We consider the case with $\gamma \ll \Omega$, allowing us to write $\eta = i\hat{\eta}$ for $\hat{\eta} \in \mathbb{R}$, and

$$\left| \cos\left(\frac{\hat{\eta} h}{2} - \operatorname{atanh}\left(\frac{h}{\hat{\eta}} \right) \right) \sqrt{1 + \frac{h^2}{\hat{\eta}^2}} \right| \leq \cosh\left(\frac{\gamma h}{2} \right).$$

The term on the left hand side of the inequality is oscillatory in h, while the term on the right hand side uniformly grows with step size. This implies that for any suitable $\gamma > 0$, we can find small pocket regions of the step size where the left hand side is sufficiently minimized in order to give stability. However, just as in the deterministic case, these island stability regions are difficult to predict the location of in a more general system.

The oscillatory cosine term is bounded between ± 1, implying that we have stability for all step sizes provided the condition

$$\sqrt{1 + \frac{h^2}{\hat{\eta}^2}} \leq \cosh\left(\frac{\gamma h}{2} \right), \tag{7.49}$$

holds. We expect there to exist stability regions for smaller values of the friction where this condition is violated, but these will be interposed between unstable regions coming from the oscillatory term. We have guaranteed stability where (7.49) holds, which we can simplify to

$$\frac{h}{\sqrt{4\Omega^2 - \gamma^2}} \leq \sinh\left(\frac{\gamma h}{2} \right).$$

For a fixed value of the friction, if we increase the step size the right hand side will grow exponentially compared to the linear growth on the left hand side, so it would seem that for large h we actually need a smaller friction in order to be stable. Hence the minimum required friction for stability can be found by considering the case of small timestep. This gives a bound

$$\frac{1}{\sqrt{4\Omega^2 - \gamma^2}} \leq \frac{\gamma}{2},$$

which gives a stability condition for all step sizes h as long as

$$\gamma \geq \Omega \sqrt{2 - 2\sqrt{1 - \Omega^{-4}}} \geq \sqrt{2}/\Omega.$$

Therefore we expect that the minimum friction value required for stability for all timesteps $h > 0$, scales with the value of inverse Ω. Unlike the deterministic case, we are able to remove linear resonance effects with a sufficiently large friction term.

We illustrate this numerically in Fig. 7.13, by applying the splitting scheme (7.48) using various timesteps and frictions. Each gridpoint marks an independent simula-